

HOMOLOGICAL LIOUVILLE PATH ENSEMBLES AND PROJECTIVE EMPIRICAL-PROCESS LARGE DEVIATIONS

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ABSTRACT. For the compact finite-horizon Lorentz family we build a parent-fold transfer bundle in which symbolic cut births are coordinate changes of one physical inverse branch. Returned-branch temporal nonintegrability, homological and collision-count periodic obstructions, a multi-block very-high-frequency estimate, and a gap-free Fourier partition yield a uniform vector–roof density and window local limit theorem.

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1. ROUND-TWELVE POSITIVE CLOSURE: PARENT-FOLD TRANSFER BUNDLES AND COMPLETE FOURIER COVERAGE

The radius interval is $\mathcal{I} = [9/20, 47/100]$. A change in a symbolic cut is treated as a change of coordinates inside one physical inverse branch, not as the creation of physical mass. The arithmetic and high-frequency arguments are rebuilt without period-one winding orbits and without an uncovered Fourier band.

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1.1. The parent-fold material graph. Near a cut birth let $r = R - R_0$ and use the signed blow-up coordinate s defined by $r = s|s|$. The physical inverse branch before cutting is a single parent operator $\mathcal{K}_R$. On the split side it is represented by $\mathcal{K}_{R,+} + \mathcal{K}_{R,-}$; on the unsplit side it remains $\mathcal{K}_R$.

Lemma 1.1 (Fold push-forward identity). *There are material charts in which*

$$\mathcal{K}_{R,+} + \mathcal{K}_{R,-} = \mathcal{K}_R$$

as distributions on the common physical test space, and $R \mapsto \mathcal{K}_R$ is C^r across R_0 . The odd split coordinate is $s\mathcal{J}_s$ with $\mathcal{J}_s$ uniformly bounded and is annihilated by physical realization at $s = 0$. Thus every physical derivative is taken from the parent operator; no Heaviside turn-on term is differentiated.

Proof. Straighten the moving singular curve and write the parent inverse integral as an integral over a fixed interval. On the split side partition that interval at the two roots of the fold equation and change variables separately on the two pieces. Adding the two integrals gives the original fixed-domain integral exactly. Differentiation in R is therefore ordinary differentiation of the parent integrand and Jacobian. Subtracting the two piece integrals gives an odd function of s ; division by s has a bounded Taylor integral representation. This proves all assertions in the common strong/weak distribution scale. $\square$

Theorem 1.2 (Uniform moving-billiard bundle). *The bulk space together with the parent-fold trace graph forms a finite atlas of Banach pairs $\mathbb{B}_R \hookrightarrow \mathbb{B}_{R,w}$ on $\mathcal{I}$. The vector-roof operators*

$$\mathcal{L}_{R,z,\zeta}h = \mathcal{L}_R(e^{z \cdot \kappa_R - \zeta \tau_R} h)$$

have a common Lasota–Yorke inequality, compact strong-to-weak injection, and a simple analytic eigenvalue on one complex neighborhood. Each material R -derivative loses one declared strong level; physical even combinations are C^r in R through every cut birth.

Proof. Finite horizon gives common flight, curvature, cone, distortion and one-step-expansion constants. Homogeneity strips make the roof multiplier summable with the same constants. Lemma 1.1 identifies all birth charts with the unsplit physical branch, so no trace-only diagonal block or peripheral mode is added. The growth lemma yields the common Lasota–Yorke estimate and compactness. A common resolvent contour and the material graph estimate give analytic source perturbation and the stated regularity loss. $\square$

1.2. Homology, collision count, and temporal nonintegrability. The normal two-cycle between neighboring disks has collision count two and roof $2(1 - 2R)$. The symmetric triangular orbit facing the center of an equilateral lattice triangle has collision count three and roof $3(1 - \sqrt{3}R)$; throughout $\mathcal{I}$ it is regular and disjoint from every other disk. These two zero-homology orbits have distinct collision counts.

Lemma 1.3 (Returned-branch UNI). *There are a common unstable interval J , two returned inverse branches $h_R^\pm : J \rightarrow J$, and $c_0 > 0$ such that their collision counts and homologies coincide while*

$$\inf_{R \in \mathcal{I}} \inf_{x \in J} \left| \partial_x (S\tau_R \circ h_R^+ - S\tau_R \circ h_R^-)(x) \right| \geq c_0.$$

The branches, their common connector, and all distortion constants vary C^2 in R .

Proof. Use the two mirror itineraries through the triangular gap, one visiting the right inward arc before the left and the other in the opposite order. The strict blocking margin $2R - \sqrt{3}/2 \geq 9/10 - \sqrt{3}/2$ keeps both itineraries away from grazing. At the symmetry point the first unequal free legs have endpoint-momentum derivatives differing by $2 \sin \beta_R$, where elementary triangle geometry gives $\sin \beta_R \geq (2R - \sqrt{3}/2)/(2 + 2\tau_+)$. Append the same number of common return iterates to both branches. Hyperbolicity bounds the derivative of the common suffix by a geometric series; choose the number of iterates so this is at most half the preceding lower bound. Continuity gives one interval J and one c_0 . $\square$

Theorem 1.4 (Joint periodic obstruction theorem). *If $u \cdot \kappa_R + vr_R + b\tau_R$ is a measurable coboundary modulo 2π , then (u, v, b) belongs to the trivial lattice character. Uniformly in R , the four coordinates $(\kappa_1, \kappa_2, r, \tau)$ have full periodic rank.*

Proof. Topological mixing and the finite connector family give regular periodic orbits with homologies e_1 and e_2 ; the closing construction stays in a fixed homogeneous core, so it is uniform in R . Close the two UNI branches with the same connector. Their homology and collision count agree, whereas their roofs differ by Lemma 1.3; subtracting the periodic identities forces $b = 0$. The normal two-cycle and triangular three-cycle then force the collision-count character to be trivial. Finally the two winding orbits force the two homology characters to be integral multiples of 2π . $\square$

1.3. A frequency partition with no gap. Set $L_n = \exp(\kappa\sqrt{n})$, where $0 < \kappa < \sqrt{c/(4A)}$ for the constants in the Dolgopyat estimate. Fourier space is split into

$$|b| \leq n^{-2/5}, \quad n^{-2/5} < |b| \leq b_0, \quad b_0 < |b| \leq L_n, \quad |b| > L_n.$$

Lemma 1.5 (Multi-block very-high-frequency estimate). *There are $\eta, c, C > 0$ such that, for $|b| > L_n$,*

$$\|\mathcal{L}_{R,iu,ib}^n\|_{\mathbb{B}_R \rightarrow \mathbb{B}_{R,w}} \leq Ce^{-cn} + C \left(\frac{C}{|b|} \right)^{\eta n}.$$

The estimate holds with every fixed central-cylinder insertion and uniformly in R and $u \in [-\pi, \pi]^2$.

Proof. The growth lemma returns a fixed fraction of every standard family to the UNI rectangle in a bounded number of iterates. Except for a set of weight Ce^{-cn} , an n -itinerary contains at least ηn disjoint returned UNI blocks. On each block use the branch coordinate from Lemma 1.3 and integrate once by parts in the phase $b(S\tau^+ - S\tau^-)$. Bounded distortion pays a uniform factor $C/|b|$. Because the blocks are disjoint, the factors multiply. Boundary terms cancel between the paired branches; homogeneity endpoints are assigned to the exponentially small bad family. A fixed insertion removes only finitely many blocks. $\square$

Theorem 1.6 (Complete Fourier estimate). *The four ranges above satisfy, respectively:*

- (i) *the analytic Gaussian expansion with cubic remainder;*
- (ii) *a uniform compact-frequency spectral gap;*
- (iii) $C|b|^A \exp\{-cn/\log(2+|b|)\} \leq Ce^{-c'\sqrt{n}};$
- (iv) *the estimate of Lemma 1.5.*

All nonsmall ranges are integrable $o(n^{-3/2})$ errors, with all required source and insertion derivatives.

Proof. The first range is analytic perturbation. The second follows from Theorem 1.4 and compactness. In the third range, $\log|b| \leq \kappa\sqrt{n}$ and the choice of κ makes the negative Dolgopyat exponent dominate $A \log|b|$. The fourth range is the preceding lemma. Integrating the resulting exponential and power tails proves the last claim. $\square$

1.4. Density and window local limits. Let $(K_n, T_n) = (S_n\kappa_R, S_n\tau_R)$ and let Σ_R be their joint covariance. Write σ_c^2 for the conditional roof variance given the homology coordinate.

Theorem 1.7 (Uniform vector-roof density LLT). *Uniformly for $k - na = O(\sqrt{n})$ and $t - n\bar{\tau} = O(\sqrt{n})$,*

$$\mathbb{P}_R(K_n = k, T_n \in dt) = \frac{e^{-\frac{1}{2}z^T \Sigma_R^{-1} z}}{(2\pi n)^{3/2} \sqrt{\det \Sigma_R}} (1 + O(n^{-\eta})) dt,$$

with the corresponding central-cylinder insertion. For a roof-sum interval $[t_n - b_n, t_n + b_n]$ the Gaussian factor is the exact conditional interval mass

$$\Phi\left(\frac{r_n + b_n}{\sqrt{n}\sigma_c}\right) - \Phi\left(\frac{r_n - b_n}{\sqrt{n}\sigma_c}\right).$$

The formula is uniform in the local, central, and saturated $b_n/\sqrt{n}$ regimes; fixed average-width windows are treated by a varying real saddle and not by this central formula.

Proof. Fourier inversion uses Theorem 1.6. The small range gives the joint Gaussian density, while the other three ranges are integrable relative errors. Integrating the density over the interval gives the exact conditional Gaussian mass and hence all three central regimes. When b_n is order n , cover the

interval by compact real-saddle charts and use Laplace's method with the analytic pressure. $\square$

Theorem 1.8 (Round-twelve A2 closure). *The compact finite-horizon Lorentz family has a uniform parent-fold transfer bundle, a complete homology-count-roof obstruction theorem, returned-branch UNI, a gap-free Fourier decomposition, and the vector-roof density and window local limits of Theorem 1.7.*

Proof. Combine Theorems 1.2, 1.4, 1.6, and 1.7. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-twelve dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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